

Department of Electrical and Computer Engineering
University of Wisconsin – Madison
ECE 552 Introduction to Computer Architecture
In-Class Exercise (09/12)

1. Assume you fetch one instruction from memory, shown in binary below. The most-significant bit (bit 31) is leftmost, and the least-significant bit (bit 0) is rightmost. Write down the instruction in **MIPS assembly syntax** (i.e., opcode, register operands).

00000001000010011000000000100100

Answer:

and \$s0, \$t0, \$t1

2. Consider the following MIPS assembly program.

```
L0: addi $s0, $s0, 0
L1: addi $s1, $s1, 1
L2: addi $s2, $s2, 2
L3: addi $s3, $s3, 3
L4: beq  $s4, $zero, _____
```

Fill in the blank assuming that the binary encoding of instruction L4 is as follows:

00010010100000001111111111111100

Answer:

L1, branch target = PC + 4 + BranchAddr = L4 + 4 + (-16) = L4 – 12 = L1

3. Consider the following MIPS assembly program. Assume that there is a bug in your machine that leads to both **bits 1 and 18** of the instruction word both being **stuck at 1** for every instruction. Rewrite this program to reflect the instructions that would actually execute on your processor due to this bug. A reminder that in MIPS, any unused bits of the instruction encoding are set to 0.

```
L0: addi $s1, $s2, 1
L1: add  $s4, $s2, $t5
L2: ori  $s1, $s1, 34
L3: sll  $s0, $t5, 8
L4: lw   $s4, 0($s0)
L5: bne  $s4, $zero, L2
```

Answer:

```
L0: addi $s5, $s2, 3
L1: sub  $s4, $s2, $t5
L2: ori  $s5, $s1, 34
L3: srl  $s0, $t5, 8
L4: lw   $s4, 2($s0)
L5: bne  $s4, $a0, L4
```